

Team Bodø

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1 Introduction

Satellite imagery plays an important role in applications such as environmental monitoring, land use analysis, agriculture, disaster response, and urban planning. However, satellite images often contain missing or corrupted regions caused by cloud cover, sensor failures, transmission errors, or occlusions. Recovering these missing areas is known as image inpainting, where the objective is to reconstruct plausible content while preserving consistency with the surrounding image.

The project topic was selected partly due to previous experience within the group with satellite image analysis and processing. Having prior familiarity with satellite imagery and its characteristics motivated us to investigate a practical problem within this domain. Unlike natural images, satellite images contain distinct patterns and structures such as fields, roads, water bodies, and urban layouts, which can present challenges for general-purpose image generation models.

Instead of developing an entirely new model, the goal of this project was to investigate whether an existing inpainting model could be adapted for satellite imagery. We used the Stable Diffusion 1.5 inpainting model as a baseline and explored the use of Low-Rank Adaptation (LoRA) to improve its performance on satellite images. LoRA provides a lightweight fine-tuning strategy that updates only a small subset of parameters rather than retraining the entire model. This approach reduces computational cost while allowing the model to adapt to image characteristics.

The objective of this project was therefore to determine whether LoRA fine-tuning could improve the quality of satellite image inpainting compared to the original Stable Diffusion 1.5 inpainting model.

2 Background

Diffusion models have recently become one of the most successful approaches for image generation tasks. These models generate images by learning a gradual denoising process, where random noise is transformed into meaningful image content over multiple steps. Stable Diffusion is a latent diffusion model that performs this process in a compressed latent space rather than directly in pixel space, significantly reducing computational requirements while maintaining image quality. The latent diffusion framework introduced by Rombach et al. demonstrated state-of-the-art performance in image synthesis and inpainting tasks (**rombach2022latentdiffusion**).

Stable Diffusion can also perform image editing tasks such as inpainting. Inpainting involves reconstructing masked regions of an image while maintaining consistency with surrounding content. The model uses information from the visible image regions and additional conditioning information to generate plausible content for the missing areas.

Although Stable Diffusion models perform well on general image generation tasks, they are primarily trained on large-scale internet image datasets. Consequently, domain-specific imagery such as satellite data may differ significantly from the training distribution. Satellite images contain unique textures, perspectives, and semantic structures compared to natural photographs, which can lead to reduced performance when using a model without adaptation (**stablediffusiongithub**).

Fine-tuning large diffusion models can be computationally expensive because of their large number of parameters. Low-Rank Adaptation (LoRA) was introduced as an efficient method for adapting large pre-trained models by learning low-rank updates to selected layers while keeping the original model weights fixed (**hu2021lora**). Instead of modifying the entire model, LoRA inserts small trainable matrices into existing layers, reducing memory usage and training time. This makes LoRA particularly suitable for projects with limited computational resources while still allowing specialization toward new domains.

In this project, LoRA was applied to the Stable Diffusion 1.5 inpainting model to adapt it toward satellite image characteristics and potentially improve reconstruction quality.

3 Dataset Preparation

Several different datasets were tested throughout the project in an attempt to adapt the Stable Diffusion 1.5 inpainting model toward satellite imagery.

Initially, two larger datasets containing thousands of satellite images were used. One dataset consisted primarily of buildings from South America, while the other focused on forested areas and trees from Poland. These datasets were used to train two separate LoRA models. However, the resulting models produced poor inpainting performance and limited visual improvement compared to the original Stable Diffusion model.

The initial experiments were performed primarily using high-detail satellite imagery obtained from sources such as Google Earth. While these images contained significantly more visual detail, they also proved difficult for the inpainting model to reconstruct accurately. Generated outputs often contained distorted textures and unrealistic structures, likely because Stable Diffusion 1.5 was originally trained on natural images rather than satellite imagery.

To simplify the reconstruction task, later experiments transitioned toward Sentinel satellite imagery, which has substantially lower spatial resolution compared to Google Earth imagery. Although the visual quality of the Sentinel images was lower, the simpler image structure made the inpainting task easier for the model. The reduced detail helped hide many of the artifacts produced during generation while still preserving the larger structural patterns present in the imagery. After the initial large-scale experiments, inspiration was taken from the DreamBooth approach (**dreambooth**), where a model can be adapted using only a few highly representative images. Based on this idea, two significantly smaller custom datasets were created manually. One dataset focused on trees, while the second focused on houses and buildings. Each dataset contained approximately 4–5 carefully selected satellite images representing the desired structures.

For all datasets, the target image resolution was kept close to 1024×1024 pixels whenever possible in order to preserve detail within the satellite imagery. Different preprocessing methods were tested throughout the experiments. For the two larger datasets, masks were created so that the model primarily trained on the target objects, such as buildings or trees, while ignoring surrounding regions. The smaller datasets were initially trained using the original unmodified images and later retrained using masked versions of the same images to test whether isolating the target objects would improve learning.

A final experiment with the smaller datasets attempted to isolate individual objects more aggressively. Instead of training on an entire image containing multiple houses or trees, each masked object within an image was treated as a separate training example. This was done to reduce the tendency of nearby structures to blend together during generation and to encourage the model to learn individual object characteristics more clearly.

4 Results and Evaluation

Model performance was evaluated using both visual inspection and Fréchet Inception Distance (FID) measurements. FID was used to compare the similarity between generated inpainted images and the target dataset distributions, where lower values indicate better performance.

Initial experiments using large datasets and high detail Google Earth imagery generally produced poor visual results. The generated outputs often contained distorted textures, unrealistic structures, and blending artifacts. This suggested that the Stable Diffusion 1.5 inpainting model struggled to generalize effectively to satellite imagery, likely because the original model was trained primarily on natural photographic images.

However, later experiments using the smaller custom house dataset produced measurable improvements after extended LoRA training. The most successful configuration used the smaller house dataset trained for approximately 3000 steps. Quantitative evaluation showed that the LoRA adapted model achieved a significantly lower FID score compared to the original base model.

[Base model FID = 224.63 LoRA model FID = 164.83]

Since lower FID values indicate better similarity to the target image distribution, this result suggests that the LoRA fine-tuning improved the model’s ability to generate satellite-like structures within masked regions. Additional testing measured the mean absolute difference between the base model outputs and the LoRA generated outputs across multiple samples:

[[10.22, 10.24, 12.33, 8.44]]

These values indicate that the LoRA model was producing noticeably different outputs compared to the original Stable Diffusion inpainting model, confirming that the finetuning process had a measurable effect on generation behavior.

The prompt used during evaluation was:

"satellite view of small houses, roof geometry, driveways, residential block"

A negative prompt was also used to suppress common artifacts:

"blurry, distorted, repeated roofs, warped perspective, low quality"

Although the quantitative metrics improved, the generated images still contained visible artifacts and inconsistencies, especially in high-detail Google Earth imagery. The model frequently struggled with fine structural details such as roof geometry, perspective consistency, and object boundaries.

To address this issue, later experiments transitioned toward lower resolution Sentinel satellite imagery. The reduced spatial detail simplified the reconstruction task and visually hid many of the artifacts produced by the model while still preserving larger structural features.

!!! INSERT SENTINEL IMAGE RESULTS AND FID COMPARISON
HERE !!!

Må compare lora der æ brukt google earth bilda av høyer resolution med
sentinel dataset direkte her. Må vis om dem improve kvalitet og FID performace

5 Discussion

Did LoRA improve performance? Where did it improve? Failure cases? Limitations

6 Conclusion